

NATIONAL AGRO-FOOD QUALITY & NMR RESEARCH LABORATORY

Accredited Analytical Testing Facility • ISO/IEC 17025:2017 Certified

BEEPROOF QUALITY ASSURANCE PROGRAM • CERTIFICATE OF ANALYSIS

CERTIFICATE NUMBER:	BP-NABL-2026-00492	STATUS:	PASS / VERIFIED
SAMPLE IDENTIFIER:	BP-2026-SUN-001 (Sundarbans Mangrove Cluster)		
TESTING PROTOCOL:	NMR Spectroscopy + Stable Isotope EA-IRMS (C4) + Physicochemical		
ISSUE DATE:	August 25, 2026		

ANALYTICAL TEST RESULTS & PHYSICOCHEMICAL PROFILE

PARAMETER / TEST	OBSERVED	SPECIFICATION / STANDARD	VERDICT
1. Moisture Content (Refractometry)	17.8%	Standard Max 20.0%	COMPLIANT
2. Pollen Purity (Floral Spectrum)	96.2%	Min 70.0% (Khalisha/Goran)	VERIFIED
3. 1H-NMR Spectroscopy Profile	PASSED	Authentic / Unadulterated	PASSED
4. C4 Sugar Adulteration (EA-IRMS)	NEGATIVE	0.0% Added Sugar (Limit <7%)	NEGATIVE
5. Hydroxymethylfurfural (HMF HPLC)	12.4 mg/kg	Max 80 mg/kg (FSSAI)	COMPLIANT
6. Diastase Enzyme Activity	18.6 DN	Min 8.0 Schade Units	COMPLIANT
7. Reducing Sugars (Fructose + Glucose)	72.4%	Min 65.0%	COMPLIANT
8. Sucrose Content	1.8%	Max 5.0%	COMPLIANT

QUALITY CERTIFICATION VERDICT: FULL CONFORMANCE (PASS)

The sample tested conforms to all FSSAI & international Codex Alimentarius standards for unadulterated raw honey.
NMR profile confirms zero exogenous sugar sirup additions (C3/C4).

CRYPTOGRAPHIC PROVENANCE ANCHOR (BEEPROOF LEDGER)

Tx Hash: 0x903bc9ed078535a77dfc47da1eb44d20eb6c1365ec40eb049d49b8c44dbe8f5

State Hash: 0xa910dc26b92a5e5e302b51a4209f35c671000773444a6d9645b38169ed0659a4

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